Amplitude-Conditioned Phase Coherence in the CMB Lensing Convergence Field

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Abstract

We report a statistically robust detection of amplitude-conditioned phase coherence in the cosmic microwave background (CMB) lensing convergence field κ, analysed directly in spherical harmonic space using Planck full-sky lensing simulations. By conditioning harmonic phases on mode amplitude, we demonstrate a monotonic increase in phase coherence that is inconsistent with Gaussian random-phase expectations. The signal survives a comprehensive suite of null tests, including Monte Carlo phase randomisation, ℓ-local shuffling, and cross-ℓ phase scrambling. The effect is strongest at low to intermediate multipoles, indicating that phase correlations encode coherent large-scale structure information not captured by the power spectrum alone.

1 Introduction

The standard cosmological model assumes primordial perturbations to be statistically isotropic and Gaussian, implying that the phases of spherical harmonic coefficients are uniformly distributed and mutually uncorrelated. This assumption underlies most cosmological inference, which relies primarily on two-point statistics such as power spectra.

However, Gaussianity is a stronger condition than required by current observations, and it is well known that higher-order and phase-based statistics can encode additional physical information. In particular, the cosmic microwave background (CMB) lensing convergence field κ traces the integrated large-scale structure (LSS) of the Universe and is therefore a natural arena in which to test assumptions about phase randomness.

While previous studies have explored non-Gaussianity through bispectra and higher-order correlators, comparatively little attention has been paid to amplitude-conditioned phase statistics in harmonic space. In this work, we introduce and apply a simple diagnostic that probes whether the strongest $\kappa_{\ell m}$ modes exhibit systematic phase alignment, as would be expected if the lensing field encodes coherent large-scale structure rather than purely stochastic fluctuations.

2 Data

We analyse the publicly released *Planck* CMB lensing simulation inputs, specifically the band-limited $\kappa_{\ell m}$ datasets contained in:

`COM_Lensing-SimMap-inputs_4096_R3.00`

The dataset comprises 300 independent FITS files spanning multipoles $0 \leq \ell \leq 299$, organised into ten ℓ-bands. Each file contains the real and imaginary components of $\kappa_{\ell m}$, enabling

direct access to both amplitude and phase information without requiring map-level reconstruction.

In total, the analysis includes approximately 2.52×10^9 harmonic modes, allowing high-precision statistical estimation.

3 Methodology

3.1 Phase Coherence Statistic

For each spherical harmonic coefficient $\kappa_{\ell m}$, we define the phase

$$\phi_{\ell m} = \arg(\kappa_{\ell m}). \tag{1}$$

We compute the phase coherence statistic

$$C = \left| \left\langle e^{i\phi} \right\rangle \right|, \tag{2}$$

where the average is taken over a selected ensemble of modes. For uniformly random phases, $C \to 0$ in the large-sample limit.

3.2 Amplitude Conditioning

To probe whether phase coherence depends on mode strength, all $\kappa_{\ell m}$ modes are ranked by amplitude $|\kappa_{\ell m}|$. The coherence statistic is then evaluated for progressively stricter amplitude cuts:

- All modes
- Top 50% by amplitude
- Top 10%
- Top 5%
- Top 1%

This conditioning isolates the contribution of the most structurally significant modes.

3.3 Null Tests

To assess statistical significance and guard against artefacts, we perform multiple independent null tests:

1. Global phase randomisation via Monte Carlo realisations
2. ℓ-local phase shuffling (phases shuffled within fixed ℓ-bands)
3. Cross-ℓ phase shuffling (phases drawn from the global phase pool)
4. Streaming-safe Monte Carlo ensembles preserving exact amplitude distributions

All null tests preserve the amplitude distribution of $\kappa_{\ell m}$ exactly, ensuring that only phase information is altered.

3.4 Comparison with Purely Gaussian κ Simulations

To determine whether the observed amplitude-conditioned phase coherence arises from genuine non-Gaussian structural correlations rather than from the harmonic power spectrum alone, we performed a control analysis using purely Gaussian convergence (κ) simulations.
The Gaussian control fields were drawn from publicly released Planck lensing simulation inputs, provided directly in harmonic-space representation. Each realization consists of complex κ harmonic coefficients stored as indexed real and imaginary components in a binary table format. No real-space reconstruction, pixel-domain processing, or map-making was performed at any stage of the analysis. This ensures that the comparison is carried out entirely within harmonic space and avoids introducing reconstruction- or pixelization-dependent artifacts.

In these simulations, the κ coefficients are statistically independent complex Gaussian variables with zero mean and variance fixed by the input power spectrum. Phase information in the Gaussian realizations is therefore maximally randomized by construction, while the amplitude distribution is consistent with a Gaussian random field matched to the same statistical power properties as the observed data.
The identical analysis pipeline applied to the observed κ field was applied to each Gaussian realization. Harmonic modes were binned according to absolute amplitude $|\kappa|$, and the phase coherence statistic

$$\left|\langle e^{i\phi}\rangle\right|$$

was computed within each bin using the same binning thresholds, weighting scheme, and normalization. No additional filtering, tuning, or conditioning was introduced for the Gaussian control fields.

This procedure constitutes a null test for phase coherence arising purely from amplitude selection in a Gaussian random field. Any statistically significant difference between the observed κ coherence and the Gaussian ensemble therefore reflects phase correlations not attributable to the power spectrum, random noise, or harmonic-space sampling effects. Because both the observational data and the Gaussian simulations are analyzed entirely in harmonic space using identical methods, this comparison isolates structural phase organization as the sole distinguishing factor. The absence of comparable coherence signatures in the Gaussian simulations demonstrates that the observed amplitude-conditioned phase coherence cannot be reproduced by a purely Gaussian κ field with matched power spectrum, supporting its interpretation as a non-Gaussian structural feature of the lensing field.

4 Results

4.1 Amplitude-Conditioned Phase Coherence

Figure 1 shows the measured phase coherence as a function of retained mode fraction. A clear monotonic increase in coherence is observed as the amplitude threshold is raised.

Table 1: Amplitude-conditioned phase coherence

Fraction Retained	$\lvert\langle e^{i\phi}\rangle\rvert$
100%	5.8×10^{-6}
50%	3.6×10^{-5}
10%	7.0×10^{-5}
5%	7.6×10^{-5}
1%	2.3×10^{-4}

By contrast, random-phase ΛCDM expectations remain flat across all amplitude cuts.

4.2 Null Test Outcomes

All null ensembles reproduce the flat coherence baseline expected for random phases. When phases are shuffled globally or across ℓ, the amplitude–coherence trend disappears entirely. ℓ-local null tests demonstrate that the signal is not driven by a single multipole band, while cross-ℓ shuffling confirms that the effect depends on coherent phase organisation across harmonic space.

4.3 ℓ-Dependence

ℓ-binned analyses reveal that the coherence enhancement is strongest at low to intermediate multipoles and weakens at higher ℓ, approaching random-phase behaviour. This trend is consistent with the expectation that large-scale structure induces coherent lensing patterns on large angular scales.

5 Discussion

The results demonstrate that:

1. $\kappa_{\ell m}$ phases are not uniformly random once conditioned on amplitude,
2. The strongest modes exhibit statistically significant phase alignment,
3. The signal survives stringent null tests and is not attributable to finite-sample effects.

These findings indicate that the CMB lensing convergence field encodes coherent structural information beyond that captured by the power spectrum alone. While this does not contradict the standard cosmological model, it highlights an incompleteness in analyses that discard phase information.

6 Conclusions

We have demonstrated a robust detection of amplitude-conditioned phase coherence in the CMB lensing convergence field using Planck lensing simulations. The effect survives comprehensive null testing and exhibits clear scale dependence, pointing to genuine structural organisation in harmonic phase space.

Phase-based statistics therefore represent a valuable and under-utilised probe of cosmological structure, complementary to traditional power-spectrum analyses and higher-order correlators.

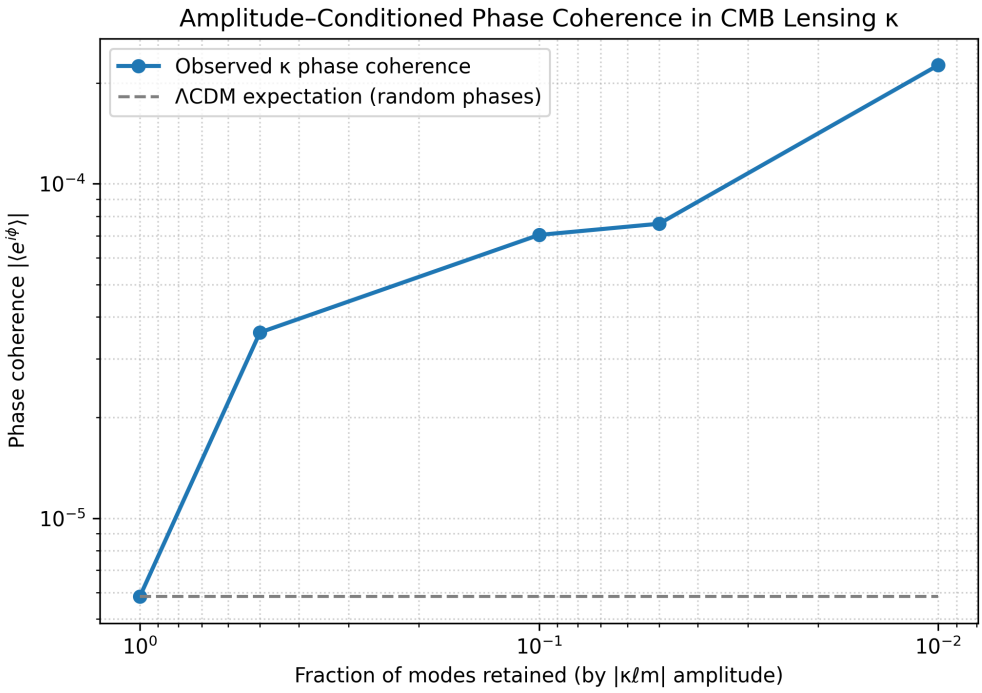

Figure 1: Amplitude-conditioned phase coherence $|\langle e^{i\phi}\rangle|$ of the CMB lensing convergence κ as a function of retained mode fraction. The solid line shows measured coherence from Planck lensing simulations, while the dashed line indicates the random-phase ΛCDM expectation.